The Chemistry Reflex: Expression → Equation Flashcard Deck

> **If you can calculate it, you have understood it.**
>
> This deck trains one reflex: **see a phrase → write the equation.** No hesitation. No "which formula?" No "do I use moles or molarity?" The word itself is the command. Memorize these mappings the way you memorized vocabulary for the SAT. When you sit down for the exam, the equations will already be on the page before you finish reading the problem.

How to Use This Deck

1. **Cover the right column.** Read the trigger phrase. Say the equation aloud. Uncover. Check.
2. **Drill by domain.** Master Stoichiometry before touching Equilibrium. Each domain is self-contained.
3. **Daily minimum:** 25 cards. Rotate domains. 213 cards — cycle through in 8–9 days.
4. **Exam simulation:** When doing practice problems, Ctrl+F this document. The goal is to need Ctrl+F less and less.

Signal system:

- **[HIGH-FREQ]** — High-frequency exam trigger. Appears in 50%+ of problems in this domain.
- **[DISTINGUISHER]** — Separates A students from B students. Often tested.
- **[MEMORIZE]** — Memorization item. A constant, a special case, or a value you just need to know.

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1. Stoichiometry — "How Much Stuff"

> **Core reflex:** "grams," "moles," "molar mass," "limiting," "yield" → convert to moles → use mole ratio → convert back.

1.1 Mole Conversions

#	Trigger Phrase	Instant Equation	Exam Note
1	"moles", "number of moles n", "how many moles"	$n = \frac{m}{M}$	m = mass (g), M = molar mass (g/mol). The universal bridge between mass and counting.
2	"molar mass M", "molecular weight"	$M = \frac{m}{n}$ (g/mol)	Sum of atomic masses from periodic table.
3	"grams to moles", "convert mass to moles", "find moles from mass"	$n = \frac{\text{mass (g)}}{M\ (\text{g/mol})}$	**Always convert to moles first.** Stoichiometry in grams is always wrong.
4	"moles to particles", "number of molecules", "number of atoms"	$N = n \times N_A$	$N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$ (Card 5).
5	"Avogadro's number N_A"	$N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$	Particles per mole.
6	"moles to volume at STP", "volume of gas at STP"	$V = n \times 22.4\ \text{L/mol}$	**STP only:** $0°\text{C}$, $1\ \text{atm}$. For any other conditions, use Card 19 ($PV = nRT$).
7	"STP"	$T = 273\ \text{K}$, $P = 1\ \text{atm}$, $V_m = 22.4\ \text{L/mol}$	Standard Temperature and Pressure. Implicit in any problem that says "at STP."

1.2 Chemical Equations & Mole Ratios

#	Trigger Phrase	Instant Equation	Exam Note
8	"balanced equation", "write the reaction", "chemical equation"	$a\text{A} + b\text{B} \rightarrow c\text{C} + d\text{D}$	Coefficients a, b, c, d are the mole ratios. **Balance before doing anything else.**

#	Trigger Phrase	Instant Equation	Exam Note
9	"mole ratio", "stoichiometric ratio", "molar ratio"	$\frac{n_\text{A}}{a} = \frac{n_\text{B}}{b} = \frac{n_\text{C}}{c}$	The bridge between any two substances in a reaction. Coefficients come from Card 8.
10	"grams of A → grams of B", "mass of product from mass of reactant"	$\text{g A} \xrightarrow{\div M_A} n_\text{A} \xrightarrow{\times (c/a)} n_\text{C} \xrightarrow{\times M_C} \text{g C}$	**Three-step path.** Never skip moles — different substances have different molar masses.
11	"limiting reagent", "limiting reactant", "which runs out first"	Divide n of each reactant by its coefficient. **Smallest result = limiting.**	The reactant that runs out first controls the theoretical yield.
12	"theoretical yield"	Mass of product from limiting reagent via mole ratio (Card 9).	Maximum possible product. Always computed from the limiting reagent.
13	"percent yield", "% yield", "percentage yield"	$\%\ \text{yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$	Always $\leq 100\%$ unless experimental error (wet product, impurity).
14	"excess reagent", "excess reactant", "in excess"	Initial mol − mol consumed (from limiting reagent × mole ratio).	The reactant that is NOT limiting. Some remains unreacted.

1.3 Empirical & Molecular Formulas

#	Trigger Phrase	Instant Equation	Exam Note
15	"empirical formula", "simplest formula"	Simplest whole-number ratio of atoms.	Method: assume $100\ \text{g}$ → mass → moles → divide all by smallest → multiply to whole numbers.

#	Trigger Phrase	Instant Equation	Exam Note
16	"molecular formula", "true formula"	$(\text{EF})_n$ where $n = \frac{M_{\text{molecular}}}{M_{\text{empirical}}}$	Molecular formula = empirical formula × integer n. Requires molar mass.
17	"percent composition", "mass percent", "% by mass"	$\%\ \text{element} = \frac{\text{mass of element in 1 mol}}{\text{molar mass of compound}} \times 100\%$	
18	"combustion analysis", "burned in oxygen"	C → CO_2, H → H_2O. From mass of CO_2 and H_2O, find mass of C and H.	Oxygen % by difference: $100\% - \%C - \%H$ (if only C, H, O present).

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2. Gases — "Things That Expand to Fill"

> **Core reflex:** "Gas," "pressure," "temperature," "volume" → $PV = nRT$. If sealed → combined gas law.

2.1 The Ideal Gas Law & Its Children

#	Trigger Phrase	Instant Equation	Exam Note
19	"ideal gas", "$PV = nRT$", "ideal gas law"	$PV = nRT$	$R = 0.08206\ \text{L atm mol}^{-1}\ \text{K}^{-1}$ (when P in atm, V in L). $R = 8.314\ \text{J mol}^{-1}\ \text{K}^{-1}$ (when P in Pa, V in m^3). **Pick R based on pressure unit.**

#	Trigger Phrase	Instant Equation	Exam Note
20	"gas constant R"	$R = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1} = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$	Two values. Match the pressure unit.
21	"sealed container", "fixed amount of gas", "n constant", "closed system"	$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$	Combined Gas Law. Use when NO gas enters or leaves. n and R cancel.
22	"Boyle's Law", "constant temperature", "isothermal"	$P_1V_1 = P_2V_2$	$P \propto 1/V$. T and n constant.
23	"Charles's Law", "constant pressure", "isobaric"	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	$V \propto T$. T MUST be in Kelvin.
24	"Gay-Lussac's Law", "constant volume", "isochoric"	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$	$P \propto T$. T MUST be in Kelvin.
25	"Avogadro's Law", "equal volumes", "T and P	$\frac{V_1}{n_1} = \frac{V_2}{n_2}$	Equal volumes of gases at same T and P contain equal moles.

#	Trigger Phrase	Instant Equation	Exam Note
	constant"		

2.2 Gas Mixtures, Effusion, & Kinetic Theory

#	Trigger Phrase	Instant Equation	Exam Note
26	"partial pressure", "Dalton's Law", "gas mixture"	$P_{\text{total}} = P_1 + P_2 + P_3 + \cdots$	Each gas in a mixture behaves as if it occupies the container alone.
27	"mole fraction X_A", "partial pressure from mole fraction"	$X_A = \frac{n_A}{n_{\text{total}}}$; $P_A = X_A P_{\text{total}}$	Partial pressure = mole fraction × total pressure.
28	"collecting gas over water", "gas collected by water displacement"	$P_{\text{dry gas}} = P_{\text{total}} - P_{\text{H}_2\text{O}}$	Subtract water vapor pressure (given or from table).
29	"Graham's Law", "effusion", "rate of effusion"	$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$	Lighter gas effuses faster. Rate $\propto 1/\sqrt{M}$.
30	"root-mean-square speed", "v_{rms}", "average speed of gas molecules"	$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$	M MUST be in **kg/mol** (divide g/mol by 1000). Lighter molecules move faster at same T.
31	"kinetic energy of a gas", "average KE per mole"	$\bar{K}_{\text{mol}} = \frac{3}{2}RT$ (per mole); $\bar{K} = \frac{3}{2}k_B T$ (per molecule)	Temperature IS average kinetic energy, up to the constant $\frac{3}{2}R$.
32	"van der Waals equation", "real gas", "non-ideal gas"	$\left(P + a\frac{n^2}{V^2}\right)(V - nb) = nRT$	a corrects for attraction; b corrects for molecular volume.

3. Thermochemistry — "Heat in Chemical Reactions"

> **Core reflex:** "Heat," "enthalpy," "ΔH" → $q = mc\Delta T$ or Hess's Law or $\Delta G = \Delta H - T\Delta S$.

3.1 Heat & Calorimetry

#	Trigger Phrase	Instant Equation	Exam Note
33	"heat", "thermal energy q ", "heat absorbed", "heat released"	$q = mc\Delta T$	c = specific heat capacity ($\mathrm{J\ g^{-1}\ K^{-1}}$). ΔT in °C or K (same size).
34	"specific heat of water c_w "	$c_w = 4.184\ \mathrm{J\ g^{-1}\ K^{-1}}$	**Memorize this.** Most common value in calorimetry.
35	"heat capacity C" (extensive)	$q = C\Delta T$ where $C = mc$	C = heat needed to raise temperature of the entire object by 1 K (or 1 °C).
36	"coffee-cup calorimeter", "constant-pressure calorimetry"	$q_{\text{system}} + q_{\text{surroundings}} = 0$	Hot loses exactly what cold gains. Constant pressure → $q = \Delta H$.
37	"bomb calorimeter", "constant-volume calorimetry"	$q_{\text{rxn}} = -C_{\text{cal}}\Delta T$	Constant volume → $q = \Delta U$. C_{cal} = heat capacity of the calorimeter assembly.
38	"molar heat capacity C_m "	$q = nC_m\Delta T$	Per mole instead of per gram.

3.2 Enthalpy

#	Trigger Phrase	Instant Equation	Exam Note
39	"enthalpy change	$\Delta H = H_{\text{products}} -$	Negative = exothermic (heat

#	Trigger Phrase	Instant Equation	Exam Note
	ΔH", "heat of reaction"	$H_{\text{reactants}}$	released). Positive = endothermic (heat absorbed).
40	"exothermic"	$\Delta H < 0$, heat is released to surroundings	Reaction vessel feels hot. Products have lower enthalpy.
41	"endothermic"	$\Delta H > 0$, heat is absorbed from surroundings	Reaction vessel feels cold. Products have higher enthalpy.
42	"Hess's Law", "add reactions", "enthalpy is a state function"	$\Delta H_{\text{target}} = \sum \Delta H_{\text{steps}}$	Path independent. Manipulate given reactions (reverse → flip sign, multiply → multiply ΔH) and add.
43	"standard enthalpy of formation ΔH_f°", "formation reaction"	$\Delta H^\circ_{\text{rxn}} = \sum n\Delta H_f^\circ(\text{products}) - \sum n\Delta H_f^\circ(\text{reactants})$	**ΔH_f° of any element in its standard state = 0.** This is the most common Hess's Law application.
44	"bond energy", "bond enthalpy", "estimate ΔH from bonds"	$\Delta H \approx \sum \text{BE(bonds broken)} - \sum \text{BE(bonds formed)}$	Breaking bonds costs energy (+). Forming bonds releases energy (−). Approximate.
45	"standard state"	$298\ \text{K}$, $1\ \text{atm}$ (or $1\ \text{bar}$), $1\ \text{M}$ for solutions	Superscript $\circ$ means standard state.

3.3 Entropy & Gibbs Free Energy

#	Trigger Phrase	Instant Equation	Exam Note
46	"entropy S", "disorder", "randomness"	$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$	Entropy of the universe always increases for spontaneous processes.
47	"standard entropy	$\Delta S^\circ =$	Same form as ΔH° from

#	Trigger Phrase	Instant Equation	Exam Note
	ΔS° from tabulated S°"	$\sum nS^\circ(\text{products}) - \sum nS^\circ(\text{reactants})$	ΔH_f°. Unlike ΔH_f°, S° of elements is NOT zero.
48	"Gibbs free energy ΔG", "free energy change"	$\Delta G = \Delta H - T\Delta S$	T MUST be in Kelvin. **Match units:** ΔS is usually J/(mol·K), ΔH in kJ/mol — convert one.
49	"spontaneous", "thermodynamically favorable"	$\Delta G < 0$	Negative ΔG = reaction proceeds forward without external input.
50	"ΔG at non-standard conditions", "ΔG and reaction quotient"	$\Delta G = \Delta G^\circ + RT \ln Q$	Q from Card 91. At equilibrium, $Q = K$ and $\Delta G = 0$.
51	"ΔG° from equilibrium constant", "relating thermo and equilibrium"	$\Delta G^\circ = -RT \ln K$	The bridge between thermodynamics and equilibrium. Large $K \rightarrow$ negative ΔG°.
52	"temperature where $\Delta G = 0$", "crossover temperature"	$T = \frac{\Delta H^\circ}{\Delta S^\circ}$ (when $\Delta G^\circ = 0$)	Above this T, spontaneity reverses. Only valid when ΔH° and ΔS° have the same sign.

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4. Atomic Structure & Periodicity — "The Blueprint of Matter"

> **Core reflex:** "Quantum number," "electron configuration," "periodic trend" → Aufbau, $n\ell m_\ell m_s$, atomic radius trend.

#	Trigger Phrase	Instant Equation	Exam Note
53	"quantum	n (principal, 1,2,3...), ℓ	Four numbers uniquely

#	Trigger Phrase	Instant Equation	Exam Note
	numbers", "four quantum numbers"	(azimuthal, 0 to $n-1$), m_ℓ (magnetic, $-\ell$ to $+\ell$), m_s (spin, $\pm\frac{1}{2}$)	identify every electron in an atom.
54	"electron configuration", "ground-state configuration"	Aufbau: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} \dots$	Fill lowest energy first. **Exceptions:** Cr $[\mathrm{Ar}]4s^1 3d^5$, Cu $[\mathrm{Ar}]4s^1 3d^{10}$ (half-filled and filled d subshell stability).
55	"Hund's Rule", "degenerate orbitals"	Fill each degenerate orbital singly with parallel spins before pairing.	"Empty bus seat rule." Maximizes unpaired electrons.
56	"Pauli Exclusion Principle"	No two electrons in an atom can have the identical set of four quantum numbers.	Each orbital holds at most 2 electrons, and they must have opposite spins.
57	"maximum electrons in shell n"	$2n^2$	$n=1$: 2, $n=2$: 8, $n=3$: 18, $n=4$: 32.
58	"orbitals in a subshell"	s: 1 orbital (2 e^-), p: 3 orbitals (6 e^-), d: 5 orbitals (10 e^-), f: 7 orbitals (14 e^-)	Each orbital holds 2 electrons.
59	"atomic radius trend", "size of atom"	Decreases → across a period. Increases ↓ down a group.	Caused by increasing Z_{eff} across a period.
60	"ionization energy trend", "energy to remove an electron"	Increases → across a period. Decreases ↓ down a group.	Opposite of atomic radius trend. Noble gases have the highest IE in each period.
61	"electronegativity trend"	Increases → across a period. Decreases ↓ down a group.	F is the most electronegative element (4.0 on Pauling

#	Trigger Phrase	Instant Equation	Exam Note
			scale).
62	"electron affinity", "energy when gaining an electron"	Generally becomes more negative (more favorable) → across a period.	Halogens have the most negative (most exothermic) electron affinities.
63	"isoelectronic", "same electron configuration"	Same number of electrons.	e.g., F^-, Ne, Na^+, Mg^{2+} all have 10 e⁻. Size decreases as nuclear charge increases.

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5. Bonding & Molecular Structure — "How Atoms Stick Together"

> **Core reflex:** "Lewis structure," "VSEPR," "hybridization" → count valence electrons → arrange → predict shape.

5.1 Lewis Structures & Formal Charge

#	Trigger Phrase	Instant Equation	Exam Note
64	"Lewis structure", "electron dot structure"	Count total valence e⁻. Form single bonds (2 e⁻ each). Distribute remaining e⁻ to complete octets.	H gets 2 e⁻ (duet). B and Be can be electron-deficient. Period 3+ can expand octet.
65	"formal charge", "FC"	$\text{FC} = (\text{valence } e^-) - (\text{lone pair } e^-) - \frac{1}{2}(\text{bonding } e^-)$	Sum of FCs = charge on species. Best structure: FC closest to zero on all atoms. Negative FC on electronegative atoms is acceptable.

#	Trigger Phrase	Instant Equation	Exam Note
66	"resonance", "resonance structures", "resonance hybrid"	Multiple valid Lewis structures differing only in electron placement.	Actual structure is a hybrid. Delocalized electrons. Bond order may be fractional.
67	"bond order" (MO theory)	$\text{BO} = \frac{N_{\text{bonding}} - N_{\text{antibonding}}}{2}$	For Lewis: BO = number of shared pairs between two atoms. Higher BO = stronger, shorter bond.

5.2 VSEPR & Hybridization

#	Trigger Phrase	Instant Equation	Exam Note
68	"VSEPR", "molecular geometry", "molecular shape"	Count **electron domains** (bonds + lone pairs) around the central atom.	2 domains: linear. 3: trigonal planar. 4: tetrahedral. 5: trigonal bipyramidal. 6: octahedral.
69	"electron-domain geometry" vs "molecular geometry"	Lone pairs occupy space but are NOT part of the molecular shape name.	NH_3: 4 domains → tetrahedral e^- geometry; 3 bonds + 1 lone pair → **trigonal pyramidal** molecular shape.
70	"hybridization", " sp, sp^2, sp^3"	Number of hybrid orbitals = number of electron domains.	2 domains → sp, 3 → sp^2, 4 → sp^3, 5 → sp^3d, 6 → sp^3d^2.
71	"bond angle"	sp: 180°, sp^2: 120°, sp^3: 109.5°	Lone pairs compress bond angles. NH_3: ~107°, H_2O: ~104.5°.

5.3 Bond Properties

#	Trigger Phrase	Instant Equation	Exam Note
72	"dipole moment μ", "polarity of a bond"	$\mu = q \times d$	Polar bonds + asymmetric geometry → molecular dipole. Symmetric molecules with polar bonds can be NONPOLAR (e.g., CO_2).
73	"polar molecule" vs "nonpolar molecule"	Has a NET dipole moment.	CO_2: polar C=O bonds but linear → dipoles cancel → nonpolar. H_2O: bent → dipoles add → polar.
74	"lattice energy", "energy of ionic solid formation"	$E \propto \frac{q_1 q_2}{r}$ (Coulomb's Law)	Larger ionic charges → larger lattice energy. Smaller ions → larger lattice energy.
75	"bond length trend", "bond energy trend"	Triple bond > Double > Single in energy. Single > Double > Triple in length.	More bonds between atoms = stronger and shorter.

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6. Chemical Kinetics — "How Fast"

> **Core reflex:** "Rate," "order," "half-life," "activation energy" → rate law → integrated rate law → Arrhenius.

6.1 Rate Laws

#	Trigger Phrase	Instant Equation	Exam Note
76	"rate law", "rate equation"	$\text{Rate} = k[\text{A}]^m[\text{B}]^n$	m and n are reaction orders — determined **experimentally,** not from the balanced equation coefficients.
77	"rate constant k ", "units of k"	0th order: M/s. 1st order: 1/s. 2nd order: 1/(M·s). 3rd order: 1/(M^2·s).	k increases with temperature (Card 84). Units depend on overall order.
78	"method of initial rates", "determine reaction order"	$\frac{\text{Rate}_2}{\text{Rate}_1} = \left(\frac{[\text{A}]_2}{[\text{A}]_1}\right)^m$	Compare two trials where only **one** reactant concentration changes. Hold all others constant.
79	"overall reaction order"	Sum of individual orders: $m + n + \cdots$	

6.2 Integrated Rate Laws — The "Which Plot Is Linear?" Test

#	Trigger Phrase	Instant Equation	Linear Plot	Half-Life
80	"zero-order reaction"	$[\text{A}] = [\text{A}]_0 - kt$	$[\text{A}]$ vs t	$t_{1/2} = \frac{[\text{A}]_0}{2k}$
81	"first-order reaction"	$\ln[\text{A}] = \ln[\text{A}]_0 - kt$	$\ln[\text{A}]$ vs t	$t_{1/2} = \frac{0.693}{k}$
82	"second-order reaction"	$\frac{1}{[\text{A}]} = \frac{1}{[\text{A}]_0} + kt$	$1/[\text{A}]$ vs t	$t_{1/2} = \frac{1}{k[\text{A}]_0}$

#	Trigger Phrase	Instant Equation	Linear Plot	Half-Life
	(one reactant)			
83	"half-life $t_{1/2}$" (general)	**1st order:** $0.693/k$ (independent of $[\mathrm{A}]_0$). **0th order:** $[\mathrm{A}]_0/(2k)$. **2nd order:** $1/(k[\mathrm{A}]_0)$.	**Only 1st-order half-life is constant.** If the problem says "half-life is always the same," the reaction is first-order.	

6.3 Temperature Dependence & Mechanisms

#	Trigger Phrase	Instant Equation	Exam Note
84	"Arrhenius equation", "temperature dependence of k"	$k = Ae^{-E_a/(RT)}$	A = frequency factor. E_a = activation energy (J/mol — match $R = 8.314$).
85	"two-point Arrhenius", "find E_a from k at two temperatures"	$\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$	T in Kelvin. E_a comes out in J/mol (divide by 1000 for kJ/mol).
86	"activation energy E_a", "energy barrier"	Minimum energy for reaction. From Arrhenius plot: slope = $-E_a/R$.	Catalyst lowers E_a but does NOT affect ΔH, ΔG, or K.
87	"rate-determining step", "slow step", "RDS"	The slowest step in a reaction mechanism. Rate law = rate law of this step.	The overall rate law comes from the molecularity of the slow step, NOT the overall reaction stoichiometry.
88	"catalyst", "catalyzed reaction"	Provides an alternate pathway with lower E_a. NOT consumed.	Does NOT change ΔH, ΔG, or K. Speeds up both forward and reverse reactions equally.

7. Chemical Equilibrium — "When Reactions Go Both Ways"

> **Core reflex:** "Equilibrium," "K_c," "K_p," "Le Chatelier" → equilibrium expression → ICE table → solve.

7.1 Equilibrium Constants

#	Trigger Phrase	Instant Equation	Exam Note
89	"equilibrium constant K_c", "equilibrium expression"	$K_c = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b}$	Products over reactants, each raised to its coefficient. **Pure solids and liquids do NOT appear in K.**
90	"K_p from K_c", "equilibrium constant in pressure"	$K_p = K_c(RT)^{\Delta n}$	$\Delta n =$ (moles gaseous products) – (moles gaseous reactants). **Count gases only.** If $\Delta n = 0$, $K_p = K_c$.
91	"reaction quotient Q", "which direction to reach equilibrium"	$Q = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b}$ (same form as K, but at any moment)	$Q < K$: forward. $Q = K$: at equilibrium. $Q > K$: reverse.
92	"large K", "small K", "position of equilibrium"	$K \gg 1$: products favored. $K \ll 1$: reactants favored.	$K \approx 1$: significant amounts of both.

7.2 ICE Tables — The Universal Equilibrium Tool

#	Trigger Phrase	Instant Equation	Exam Note
93	"ICE table", "equilibrium calculation"	**I**nitial → **C**hange → **E**quilibrium	Row 1: initial concentrations (M). Row 2: change ($-ax$, $-bx$, $+cx$, $+dx$) using a variable x. Row 3: sum = equilibrium concentrations.
94	"small-K approximation", "neglect x", "5% rule"	If K is small, drop x from subtraction: $[\mathrm{HA}]_0 - x \approx [\mathrm{HA}]_0$.	**Must verify:** $x/[\mathrm{initial}]_0 \times 100\% < 5\%$. If not, solve the quadratic.
95	"equilibrium concentration from ICE"	$[\mathrm{A}]_{\mathrm{eq}} = [\mathrm{A}]_0 + \mathrm{change}$ (change may be negative)	x found by solving the K expression using the "E" row of the ICE table.

7.3 Le Chatelier's Principle

#	Trigger Phrase	Instant Equation	Exam Note
96	"Le Chatelier's Principle", "shift equilibrium", "stress on equilibrium"	System shifts to **relieve** the applied stress.	Three stresses: concentration, pressure/ volume, temperature.
97	"add reactant" / "add product"	Shifts **away** from the added substance.	Add A → shift right (consumes A). Add C → shift left (consumes C).
98	"increase pressure" / "decrease volume" (gaseous equilibrium)	Shift toward the side with **fewer moles of gas.**	Only affects systems with $\Delta n \neq 0$. Adding inert gas at constant volume: NO shift.
99	"increase	Exothermic ($\Delta H < 0$): ↑	**Only temperature changes**

#	Trigger Phrase	Instant Equation	Exam Note
	temperature", "decrease temperature"	T shifts LEFT. Endothermic ($\Delta H > 0$): ↑ T shifts RIGHT.	K. Concentration and pressure changes do NOT change K.
100	"add a catalyst at equilibrium"	**No shift.** Speeds up forward and reverse equally.	K unchanged. Equilibrium reached faster, but composition is identical.

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8. Acids & Bases — "Proton Transfer"

> **Core reflex:** "pH," "acid," "base," "buffer," "titration" → K_a, K_b, K_w, Henderson-Hasselbalch, equivalence point.

8.1 pH, pOH, and Dissociation Constants

#	Trigger Phrase	Instant Equation	Exam Note
101	"pH", "find the pH", "calculate pH"	$\text{pH} = -\log[\text{H}^+]$	Reverse: $[\text{H}^+] = 10^{-\text{pH}}$.
102	"pOH", "find the pOH"	$\text{pOH} = -\log[\text{OH}^-]$	$\text{pH} + \text{pOH} = 14.00$ at $25°\text{C}$.
103	"K_w", "ion product of water", "autoionization of water"	$K_w = [\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$ at $25°\text{C}$	K_w increases with temperature. At $25°\text{C}$, pure water: $[\text{H}^+] = [\text{OH}^-] = 1.0 \times 10^{-7}\ \text{M}$, pH = 7.00.
104	"acid dissociation constant K_a", "strength of a weak acid"	$K_a = \dfrac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$	Larger K_a = stronger acid. $\text{p}K_a = -\log K_a$.
105	"base dissociation	$K_b =$	For a conjugate acid-base pair:

#	Trigger Phrase	Instant Equation	Exam Note
	constant K_b", "strength of a weak base"	$\frac{[\mathrm{BH}^+][\mathrm{OH}^-]}{[\mathrm{B}]}$	$K_a \times K_b = K_w$.
106	"percent ionization", "% dissociation"	$\% \text{ ionization} = \frac{[\mathrm{H}^+]_{\mathrm{eq}}}{[\mathrm{HA}]_0} \times 100\%$	Weak acids: % ionization **increases** with dilution (contrary to intuition).

8.2 Strong vs Weak — The Decision Tree

#	Trigger Phrase	Instant Equation	Exam Note
107	"strong acid" (list)	HCl, HBr, HI, HNO_3, H_2SO_4 (first H^+ only), $HClO_4$, $HClO_3$	**Complete dissociation.** $[\mathrm{H}^+] = [\mathrm{HA}]_0$. No ICE table needed.
108	"strong base" (list)	Group 1 hydroxides: LiOH, NaOH, KOH, RbOH, CsOH. Also $Ca(OH)_2$, $Sr(OH)_2$, $Ba(OH)_2$.	**Complete dissociation.** For $Ca(OH)_2$: $[\mathrm{OH}^-] = 2 \times [\mathrm{Ca(OH)_2}]_0$.
109	"weak acid pH", "pH of a weak acid solution"	$[\mathrm{H}^+] = \sqrt{K_a[\mathrm{HA}]_0}$	Valid ONLY when $K_a \ll [\mathrm{HA}]_0$ and 5% rule passes (Card 94). Derived from ICE table.
110	"weak base pH", "pH of a weak base solution"	$[\mathrm{OH}^-] = \sqrt{K_b[\mathrm{B}]_0}$; then $\mathrm{pH} = 14 - \mathrm{pOH}$	Same logic as Card 109, but for bases. Convert to pH at the end.

8.3 Buffers & Titrations

#	Trigger Phrase	Instant Equation	Exam Note
111	"buffer", "buffer solution", "buffered"	A solution containing a weak acid and its conjugate base (or weak base and its conjugate acid) in significant amounts.	Resists pH change when small amounts of strong acid or base are added.
112	"Henderson-Hasselbalch equation", "buffer pH"	$\mathrm{pH} = \mathrm{p}K_a + \log \frac{[\mathrm{A}^-]}{[\mathrm{HA}]}$	$[\mathrm{A}^-]$ and $[\mathrm{HA}]$ are equilibrium concentrations (≈ initial for buffers). Ratio must be 0.1–10.
113	"buffer capacity", "maximum buffer capacity"	Maximum when $[\mathrm{A}^-] = [\mathrm{HA}]$, giving $\mathrm{pH} = \mathrm{p}K_a$.	
114	"equivalence point", "endpoint of titration"	Moles of acid = moles of base added (accounting for stoichiometry).	$\mathrm{pH} \neq 7$ for weak acid/ strong base or weak base/ strong acid titrations.
115	"half-equivalence point", "halfway to equivalence"	$\mathrm{pH} = \mathrm{p}K_a$ (for titration of weak acid with strong base).	Exactly half of the weak acid has been converted to its conjugate base.
116	"titration curve", "pH curve"	Plot of pH vs volume of titrant added. Shape depends on acid and base strengths.	Strong acid + strong base: pH 7 at equivalence. Weak acid + strong base: pH > 7 at equivalence.
117	"indicator",	Weak acid with different colors for	Choose indicator with

#	Trigger Phrase	Instant Equation	Exam Note
	"acid-base indicator"	HA and A^- forms.	$\mathrm{p}K_a \approx$ pH at the equivalence point.
118	"polyprotic acid", "diprotic acid", "triprotic acid"	Multiple K_a values: $K_{a1} \gg K_{a2} \gg K_{a3}$. Each successive H^+ is harder to remove.	Multiple equivalence points in titration curve.
119	"salt hydrolysis", "pH of a salt solution"	Conjugate base of weak acid → basic. Conjugate acid of weak base → acidic.	Salt of strong acid + strong base → neutral (pH 7).
120	"amphoteric", "amphiprotic"	Can act as either an acid or a base.	e.g., H_2O, HCO_3^-, $H_2PO_4^-$, HPO_4^{2-}.

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9. Electrochemistry — "Electrons in Motion"

> **Core reflex:** "Redox," "cell potential," "electrolysis" → half-reactions → $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ → Nernst.

9.1 Redox Fundamentals

#	Trigger Phrase	Instant Equation	Exam Note
121	"oxidation", "oxidized", "oxidation half-reaction"	Loss of electrons. Oxidation number **increases.**	OIL RIG: Oxidation Is Loss, Reduction Is Gain. Occurs at the **anode.**

#	Trigger Phrase	Instant Equation	Exam Note
122	"reduction", "reduced", "reduction half-reaction"	Gain of electrons. Oxidation number **decreases.**	Occurs at the **cathode.**
123	"oxidation number", "oxidation state", "assign oxidation numbers"	Priority: F = −1 always. O = −2 (except peroxides −1, OF_2 +2). H = +1 with nonmetals (except metal hydrides −1). Sum = overall charge.	Element alone = 0. Monatomic ion = its charge.
124	"balance redox reaction" (acidic)	Half-reaction method: balance atoms, add H_2O for missing O, add H^+ for missing H, add e^- for charge. Equalize e^- and add half-reactions.	For basic solution: add OH^- to both sides to neutralize H^+, forming H_2O.

9.2 Cell Potentials

#	Trigger Phrase	Instant Equation	Exam Note
125	"standard cell potential E°_{cell}", "voltage of a galvanic cell"	$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$	Cathode = reduction (higher E°). Anode = oxidation (lower E°). **Use reduction potentials as given; do NOT flip the sign.**
126	"spontaneous redox", "galvanic cell", "voltaic cell"	$E^\circ_{\text{cell}} > 0 \rightarrow$ spontaneous. $E^\circ_{\text{cell}} < 0 \rightarrow$ nonspontaneous (electrolytic).	Positive cell potential = reaction produces electrical energy.
127	"ΔG° from E°", "free energy and cell potential"	$\Delta G^\circ = -nFE^\circ$	n = moles of e^- transferred. $F = 96{,}485\ \text{C/mol e}^-$ (Card 129).
128	"Nernst	$E = E^\circ -$	At 25°C: $E = E^\circ -$

#	Trigger Phrase	Instant Equation	Exam Note
	equation", "cell potential at non-standard conditions"	$\frac{RT}{nF}\ln Q$	$\frac{0.0592}{n}\log_{10} Q$. Use $\log_{10}$, not $\ln$, in the 0.0592 form.
129	"Faraday's constant F"	$F = 96{,}485\ \mathrm{C/mol\ e^-}$	Charge of one mole of electrons.

9.3 Electrolysis — Driving Non-Spontaneous Reactions

#	Trigger Phrase	Instant Equation	Exam Note
130	"electrolysis", "electrolytic cell"	An external voltage forces a nonspontaneous redox reaction.	$E^\circ_{\text{cell}} < 0$; external V must exceed $
131	"Faraday's Law of Electrolysis", "mass deposited during electrolysis"	$m = \frac{I \times t \times M}{n \times F}$	I = current (A), t = time (s), M = molar mass of product (g/mol), n = moles of e^- per mole of product.
132	"charge from current and time"	$Q = I \times t$ (coulombs)	Q/F = moles of electrons passed.

10. Solutions & Colligative Properties — "What Dissolved Particles Do"

> **Core reflex:** "Concentration," "molarity," "molality," "colligative" → $M = n/V$, $\Delta T = iKm$, $\Pi = iMRT$.

10.1 Concentration Units

#	Trigger Phrase	Instant Equation	Exam Note
133	"molarity M", "molar concentration"	$M = \frac{n_{\text{solute}}}{V_{\text{solution (L)}}}$	Denominator is volume of the **final solution**, NOT volume of solvent. M changes with temperature.
134	"molality m", "molal concentration"	$m = \frac{n_{\text{solute}}}{\text{kg of solvent}}$	Temperature-independent. Used in colligative property calculations.
135	"mass percent", "% by mass", "weight percent"	$\%\ \text{mass} = \frac{\text{mass solute}}{\text{mass solution}} \times 100\%$	
136	"mole fraction X_A"	$X_A = \frac{n_A}{n_{\text{total}}}$	Sum of all mole fractions in a mixture = 1.
137	"dilution", "dilute a solution", "add solvent"	$M_1V_1 = M_2V_2$	Moles of solute remain constant. Only solvent is added.
138	"ppm", "ppb", "parts per million"	$\text{ppm} = \frac{\text{mass solute}}{\text{mass solution}} \times 10^6$	For very dilute solutions. $1\ \text{ppm} \approx 1\ \text{mg/L}$ for aqueous solutions.

10.2 Colligative Properties — Depend Only on Number of Particles

#	Trigger Phrase	Instant Equation	Exam Note
139	"colligative properties"	Depend on the NUMBER of solute particles, not their identity.	Four: vapor pressure lowering, boiling point elevation, freezing point depression, osmotic pressure.

#	Trigger Phrase	Instant Equation	Exam Note
140	"Raoult's Law", "vapor pressure of a solution"	$P_A = X_A P_A^\circ$	Vapor pressure of solvent above solution. For nonvolatile solute: $\Delta P = X_{\text{solute}} P_{\text{solvent}}^\circ$.
141	"boiling point elevation", "ΔT_b"	$\Delta T_b = iK_b m$	i = van't Hoff factor. K_b = ebullioscopic constant of the solvent.
142	"freezing point depression", "ΔT_f"	$\Delta T_f = iK_f m$	K_f = cryoscopic constant.
143	"osmotic pressure Π"	$\Pi = iMRT$	M = molarity of solute. $R = 0.08206\ \text{L atm mol}^{-1}\ \text{K}^{-1}$.
144	"van't Hoff factor i"	Number of ions per formula unit upon dissolution.	Ideal: NaCl → 2, $CaCl_2$ → 3. Covalent solutes (sugar): $i = 1$. Real solutions: slightly less due to ion pairing.

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11. Nuclear Chemistry — "The Nucleus"

> **Core reflex:** "Half-life," "decay," "binding energy" → $N = N_0 e^{-\lambda t}$, $E = \Delta mc^2$.

#	Trigger Phrase	Instant Equation	Exam Note
145	"radioactive decay", "exponential decay"	$N = N_0 e^{-\lambda t}$	λ = decay constant. Same mathematics as first-order kinetics (Card 81).

#	Trigger Phrase	Instant Equation	Exam Note
146	"half-life $t_{1/2}$", "nuclear half-life"	$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$	After n half-lives: fraction remaining $=$ $(1/2)^n$.
147	"activity A", "decay rate"	$A = \lambda N$	Units: becquerel (Bq) = 1 decay/s. Curie (Ci) = 3.7×10^{10} Bq.
148	"alpha decay", "α particle emission"	${}^{A}_{Z}\mathrm{X} \to {}^{A-4}_{Z-2}\mathrm{Y} + {}^{4}_{2}\mathrm{He}$	Mass number −4, atomic number −2.
149	"beta decay (β^-)", "beta minus decay"	${}^{A}_{Z}\mathrm{X} \to {}^{A}_{Z+1}\mathrm{Y} + {}^{0}_{-1}\mathrm{e} + \bar{\nu}$	Neutron → proton + electron. Atomic number **+1**, mass number unchanged.
150	"positron emission (β^+)"	${}^{A}_{Z}\mathrm{X} \to {}^{A}_{Z-1}\mathrm{Y} + {}^{0}_{+1}\mathrm{e} + \nu$	Proton → neutron + positron. Atomic number **−1**.
151	"electron capture"	${}^{A}_{Z}\mathrm{X} + {}^{0}_{-1}\mathrm{e} \to {}^{A}_{Z-1}\mathrm{Y} + \nu$	Inner-shell electron captured by nucleus. Atomic number −1.
152	"gamma emission", "γ decay"	Excited nucleus → ground state + γ photon.	No change in A or Z. Often accompanies α or β decay.
153	"nuclear binding energy", "mass defect"	$E_B = \Delta mc^2$	Δm = (mass of nucleons) − (mass of nucleus). The "missing mass" is converted to binding energy.
154	"mass defect"	Mass of nucleus < sum of masses of its protons and neutrons.	
155	"binding	E_B/A; maximum at	Determines nuclear stability. Fusion of

#	Trigger Phrase	Instant Equation	Exam Note
	energy per nucleon"	Fe-56.	light nuclei and fission of heavy nuclei both release energy.

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12. Laboratory & Measurement — "Numbers That Mean Something"

#	Trigger Phrase	Instant Equation	Exam Note
156	"significant figures", "sig figs", "precision"	Mult/div: **fewest sig figs.** Add/sub: **fewest decimal places.**	Exact numbers (counted objects, defined conversions) have infinite sig figs.
157	"percent error", "% error"	$%;\text{error} = \dfrac{	\text{experimental} - \text{accepted}
158	"density ρ"	$\rho = \dfrac{m}{V}$	Intensive property. $\rho_{\text{water}} = 1.00\ \text{g/mL}$ at $4°\text{C}$.
159	"Beer-Lambert Law", "absorbance", "spectrophotometry"	$A = \varepsilon bc$	A = absorbance, ε = molar absorptivity, b = path length (cm), c = concentration (M).

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13. Cross-Domain Bridges — "When One Calculation Feeds Another"

> **These are not new equations — they are problem architectures.** Each bridge card describes a two-domain chain. Domain A produces a variable (the bridge), which becomes the input to Domain B.

13.1 Stoichiometry → Gas / Thermo / Equilibrium

#	Scenario	Domain A → Bridge → Domain B	Olympiad Insight
160	Mass of reactant → volume of gas	Stoich (g → mol) → n_{gas} → $V = nRT/P$ (Card 19) or $V = n \times 22.4$ at STP (Card 6)	Most common two-domain problem on the AP exam.
161	Limiting reagent → gas pressure	Find limiting reagent (Card 11) → n_{gas} → $P = nRT/V$ (Card 19)	The limiting reagent controls how much gas is produced.
162	Combustion → heat → temperature change	Stoich (g → mol) → ΔH → q → $q = mc\Delta T$ (Card 33) → ΔT	Three-domain chain. Extremely common in lab-based problems.
163	Initial amounts → equilibrium amounts	ICE table (Card 93) using initial n or M → x from K → mass or concentration of each species at equilibrium	The x from the ICE table is the bridge to everything else.

13.2 Thermo → Equilibrium / Electrochemistry

#	Scenario	Domain A → Bridge → Domain B	Olympiad Insight
164	ΔG° from thermo data → equilibrium K	$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ (Card 48) → ΔG° → $\Delta G^\circ = -RT \ln K$ (Card 51) → K	The bridge from thermodynamics to equilibrium.
165	ΔG° → cell potential	ΔG° (from Card 48 or 164) → $\Delta G^\circ = -nFE^\circ$ (Card 127) → E°	The bridge from thermodynamics to electrochemistry.
166	Cell potential → equilibrium K	E°_{cell} (Card 125) → $E^\circ = \frac{RT}{nF} \ln K$ → K	Large E°_{cell} → large K.

13.3 Kinetics → Nuclear / Mechanisms

#	Scenario	Domain A → Bridge → Domain B	Olympiad Insight
167	First-order kinetics → half-life (chemical or nuclear)	$\ln[A] = \ln[A]_0 - kt$ (Card 81) → k → $t_{1/2} = 0.693/k$ (Card 83)	**Chemical kinetics and nuclear decay use identical mathematics.** Card 146 is the nuclear twin.
168	Radioactive dating	$N = N_0 e^{-\lambda t}$ (Card 145) → t → age of sample	Solve for t given N/N_0 and λ.

13.4 Acid-Base → Equilibrium → Buffer Design

#	Scenario	Domain A → Bridge → Domain B	Olympiad Insight
169	K_a → buffer pH	$\mathrm{p}K_a = -\log K_a$ (Card 104) → $\mathrm{p}K_a$ → $\mathrm{pH} = \mathrm{p}K_a + \log([\mathrm{A}^-]/[\mathrm{HA}])$ (Card 112)	Henderson-Hasselbalch requires $\mathrm{p}K_a$.
170	Titration curve → K_a determination	Half-equivalence: $\mathrm{pH} = \mathrm{p}K_a$ (Card 115) → $\mathrm{p}K_a$ → $K_a = 10^{-\mathrm{p}K_a}$	The experimental method for determining K_a of an unknown weak acid.

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14. Organic Chemistry — "The Language of Carbon" (Honors-Level Essentials)

> **Core reflex:** "Functional group," "isomer," "reaction type" → name → structure → product.

#	Trigger Phrase	Instant Equation	Exam Note
171	"alkane"	C_nH_{2n+2}, single bonds, -ane	Methane, ethane, propane, butane, pentane…
172	"alkene"	C_nH_{2n}, contains C=C, -ene	
173	"alkyne"	C_nH_{2n-2}, contains C≡C, -yne	
174	"alcohol", "hydroxyl group"	R–OH, -ol suffix	
175	"carboxylic acid", "organic acid"	R–COOH, -oic acid	
176	"ester"	R–COO–R', from alcohol + carboxylic acid	Fragrant. -oate suffix.
177	"amine"	$R{-}NH_2$ (1°), R_2NH (2°), R_3N (3°)	-amine suffix. Basic (lone pair on N).
178	"aldehyde"	R–CHO, -al suffix	Terminal carbonyl group.
179	"ketone"	R–CO–R', -one suffix	Internal carbonyl group.
180	"ether"	R–O–R'	
181	"isomer", "structural isomer", "stereoisomer"	Same molecular formula, different structure (structural) or spatial arrangement (stereo).	Enantiomers = nonsuperimposable mirror images (chiral center, usually C with 4 different groups).
182	"addition reaction"	Alkene + XY → product (across the double bond)	π bond breaks; two new σ bonds form.
183	"substitution reaction",	$R{-}X + Nu^- \rightarrow R{-}Nu + X^-$	SN2: one-step, backside attack, inversion. SN1: two-step,

#	Trigger Phrase	Instant Equation	Exam Note
	"SN1", "SN2"		carbocation intermediate, racemization.
184	"elimination reaction"	R–CH_2–CH_2–X → R–CH=CH_2 + HX	Opposite of addition. Forms a double bond.
185	"oxidation of alcohols"	1° alcohol → aldehyde → carboxylic acid. 2° alcohol → ketone. 3° alcohol → no reaction.	Common oxidizing agents: CrO_3, $KMnO_4$, PCC (stops at aldehyde).

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15. The Implicit Information Catalog — "What the Problem Says Without Saying"

> **These phrases hide a value. If you miss them, you cannot solve the problem.**

#	Trigger Phrase	Hidden Value	Why It Matters
186	"STP"	$T = 273\ \mathrm{K}$, $P = 1\ \mathrm{atm}$, $V_m = 22.4\ \mathrm{L/mol}$	Gas stoichiometry at STP does not require $PV = nRT$.
187	"standard state" / "°"	$298\ \mathrm{K}$, $1\ \mathrm{atm}$, $1\ \mathrm{M}$; ΔH_f° of elements = 0	Required for any ΔH°, ΔG°, S°, E° calculation.
188	"excess" (reagent)	This reactant is NOT limiting. Its exact amount is irrelevant to theoretical yield.	You must still identify the limiting reagent among the other reactants.
189	"strong acid" (by name)	$[\mathrm{H}^+] = [\mathrm{HA}]_0$ — complete dissociation. No ICE table.	If you draw an ICE table for a strong acid, you are wasting time.

#	Trigger Phrase	Hidden Value	Why It Matters
190	"strong base" (by name)	$[\mathrm{OH^-}] = [\mathrm{B}]_0 \times$ (# of OH^- per formula).	
191	"ideal gas" (stated or implied)	Obeys $PV = nRT$. No intermolecular forces. Point particles.	All gas law problems assume ideality unless "van der Waals" or "real gas" is stated.
192	"at equilibrium"	$Q = K$, $\Delta G = 0$, forward rate = reverse rate.	Three equivalent statements. When any one is given, the other two are automatically true.
193	"first-order" (kinetics)	$t_{1/2} = 0.693/k$ — half-life is **constant**, independent of $[\mathrm{A}]_0$.	Most common order on the AP exam. Linear plot: $\ln[\mathrm{A}]$ vs t.
194	"catalyst present"	E_a is lowered. ΔH, ΔG, and K are **unchanged.**	Catalyst changes the path, not the thermodynamic endpoints.
195	"monoprotic acid"	One ionizable H^+ per molecule.	Equivalence: 1 mol base neutralizes 1 mol acid.
196	"diprotic acid"	Two ionizable H^+. $K_{a1} \gg K_{a2}$. Two equivalence points.	First equivalence: $H_2A \rightarrow HA^-$. Second: $HA^- \rightarrow A^{2-}$.

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16. Unit Conversion Reflexes — "Instant Conversions Under Pressure"

#	From	To	Multiply By
197	°C	K	$T_K = T_C + 273.15$
198	atm	Pa	$1\ \mathrm{atm} = 101\,325\ \mathrm{Pa}$

#	From	To	Multiply By
199	atm	torr / mmHg	$1 \text{ atm} = 760 \text{ torr}$
200	L	m^3	$1 \text{ L} = 10^{-3} \text{ m}^3$
201	mL	L (and cm^3)	$1 \text{ mL} = 10^{-3} \text{ L} = 1 \text{ cm}^3$
202	J	cal	$1 \text{ cal} = 4.184 \text{ J}$
203	kJ	J	Multiply by 1000
204	eV	J	$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$
205	eV/particle	kJ/mol	Multiply by 96.485
206	amu (u)	kg	$1 \text{ u} = 1.661 \times 10^{-27} \text{ kg}$
207	amu (u)	MeV/c^2	$1 \text{ u} = 931.5 \text{ MeV}/c^2$
208	M (molar)	mol/L	$1 \text{ M} \equiv 1 \text{ mol/L}$
209	Å (angstrom)	m	$1 \text{ Å} = 10^{-10} \text{ m}$
210	nm	m	$1 \text{ nm} = 10^{-9} \text{ m}$
211	pm	m	$1 \text{ pm} = 10^{-12} \text{ m}$
212	g/cm^3	kg/m^3	Multiply by 1000
213	g/mol	kg/mol	Divide by 1000

Quick Domain Lookup

Domain	Cards	Signature Trigger Words
1. Stoichiometry	1–18	moles, molar mass, limiting, percent yield, empirical, combustion
2. Gases	19–32	$PV = nRT$, STP, Dalton, Graham, v_{rms}, van der

Domain	Cards	Signature Trigger Words
		Waals
3. Thermochemistry	33–52	heat, enthalpy, Hess, calorimetry, entropy, Gibbs, spontaneous
4. Atomic Structure	53–63	quantum number, electron config, Aufbau, periodic trend, isoelectronic
5. Bonding	64–75	Lewis, VSEPR, hybridization, formal charge, dipole, lattice energy
6. Kinetics	76–88	rate law, order, integrated rate law, half-life, Arrhenius, catalyst
7. Equilibrium	89–100	K_c, K_p, ICE table, Q, Le Chatelier, shift
8. Acids & Bases	101–120	pH, K_a, K_b, buffer, Henderson-Hasselbalch, titration, equivalence
9. Electrochemistry	121–132	redox, E°, Nernst, electrolysis, Faraday
10. Solutions	133–144	molarity, molality, dilution, colligative, Raoult, osmotic pressure
11. Nuclear	145–155	half-life, decay, alpha, beta, binding energy, mass defect
12. Laboratory	156–159	sig figs, % error, density, Beer-Lambert
13. Cross-Domain Bridges	160–170	stoich→gas, combustion→heat, $\Delta G^\circ \to K$, $K \to E^\circ$, titration→K_a
14. Organic	171–185	alkane, functional group, isomer, addition, substitution, elimination
15. Implicit Information	186–196	STP, standard state, excess, strong acid, at equilibrium, first-order
16. Unit Conversions	197–213	$^\circ$C→K, atm→Pa, J→cal, eV→kJ/mol, u→MeV/c^2

17. Quick Diagnostic Triggers — "Test Yourself in 60 Seconds"

> **Read each trigger phrase. If you cannot instantly write the equation, go back and drill that domain.**

Trigger Phrase	You Should Instantly Write	Card
"moles from grams"	$n = m/M$	1
"limiting reagent"	Divide n by coefficient; smallest wins	11
"percent yield"	(actual/theoretical) × 100%	13
"ideal gas law"	$PV = nRT$	19
"sealed container, heated"	$P_1V_1/T_1 = P_2V_2/T_2$	21
"Dalton's Law"	$P_{\text{total}} = P_1 + P_2 + \cdots$	26
"Graham's Law"	$r_1/r_2 = \sqrt{M_2/M_1}$	29
"heat" (with ΔT)	$q = mc\Delta T$	33
"Hess's Law"	$\Delta H_{\text{rxn}} = \sum \Delta H_{\text{steps}}$	42
"ΔH° from ΔH_f°"	$\sum n\Delta H_f^\circ$(products) – $\sum n\Delta H_f^\circ$(reactants)	43
"Gibbs free energy"	$\Delta G = \Delta H - T\Delta S$	48
"rate law"	Rate = $k[\mathrm{A}]^m[\mathrm{B}]^n$	76
"first-order half-life"	$t_{1/2} = 0.693/k$	81, 83
"Arrhenius equation"	$k = Ae^{-E_a/(RT)}$	84
"equilibrium constant"	$K_c = [\mathrm{C}]^c[\mathrm{D}]^d/([\mathrm{A}]^a[\mathrm{B}]^b)$	89
"ICE table"	Initial → Change ($\pm x$, with coefficients) → Equilibrium	93

Trigger Phrase	You Should Instantly Write	Card
"Le Chatelier — increase T"	Shift away from heat (endo → right, exo → left)	99
"pH"	$\mathrm{pH} = -\log[\mathrm{H}^+]$	101
"Henderson-Hasselbalch"	$\mathrm{pH} = \mathrm{p}K_a + \log([\mathrm{A}^-]/[\mathrm{HA}])$	112
"equivalence point"	mol acid = mol base (accounting for stoichiometry)	114
"half-equivalence point"	$\mathrm{pH} = \mathrm{p}K_a$	115
"E°_{cell}"	$E^\circ_{\mathrm{cathode}} - E^\circ_{\mathrm{anode}}$	125
"Nernst equation at 25°C"	$E = E^\circ - (0.0592/n)\log Q$	128
"Faraday's Law of Electrolysis"	$m = ItM/(nF)$	131
"dilution"	$M_1V_1 = M_2V_2$	137
"boiling point elevation"	$\Delta T_b = iK_bm$	141
"freezing point depression"	$\Delta T_f = iK_fm$	142
"osmotic pressure"	$\Pi = iMRT$	143
"radioactive half-life"	$t_{1/2} = 0.693/\lambda$	146

> **The exam is won before you enter the room.** Every equation on this page is a tool. Your job is to know which tool to grab the instant you see the trigger phrase. Drill these cards until the response is automatic. When you read "limiting reagent," your hand should already be dividing moles by coefficients. When you read "equilibrium," an ICE table should already be forming.
>
> You don't need to see the whole solution. You just need to start the right equation. The algebra will take you the rest of the way.
>
> **If you can calculate it, you have understood it.**